

BARUUNKHARAA, Mongolia

WMO: 442410

Lat: 48.9133N

Lon: 106.0897E

Elev: 807

StdP: 92.00

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.8	-34.4	-39.2	0.1	-36.6	-36.7	0.1	-34.2	7.9	-14.3	6.6	-16.3	0.9	180	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	32.6	18.6	30.5	18.1	28.5	17.4	21.5	28.8	20.0	27.0	18.9	25.4	2.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	15.4	24.9	17.7	14.0	23.2	16.6	13.0	22.1	66.9	29.2	61.4	27.2	57.4	25.3	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.3	6.2	-38.1	36.6	3.4	2.6	-40.5	38.5	-42.5	40.0	-44.4	41.4	-46.9	43.3		
			-38.2	22.8	3.4	2.2	-40.6	24.4	-42.6	25.7	-44.5	26.9	-46.9	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.6	-24.4	-19.0	-6.1	4.5	11.4	18.1	20.8	18.4	11.7	1.9	-10.6	-20.6
	DBStd	16.51	5.53	6.63	7.08	5.12	5.11	4.41	3.44	3.70	4.75	5.02	7.22	5.83
	HDD10.0	4453	1065	811	500	175	44	1	0	0	36	253	618	950
	HDD18.3	6667	1323	1044	758	414	222	57	13	46	204	510	868	1208
	CDD10.0	1029	0	0	0	12	88	244	336	261	86	2	0	0
	CDD18.3	201	0	0	0	0	7	51	92	47	4	0	0	0
	CDH23.3	2407	0	0	0	7	177	684	962	501	77	0	0	0
	CDH26.7	873	0	0	0	1	51	261	374	172	14	0	0	0
Wind	WSAvg	2.1	1.4	1.6	2.3	2.8	2.8	2.5	2.0	2.0	2.2	2.0	1.8	1.5
Precipitation	PrecAvg	285	3	2	4	9	22	52	70	65	32	10	6	5
	PrecMax	473	26	13	29	31	72	155	154	159	69	47	22	20
	PrecMin	134	0	0	0	0	0	0	6	1	0	0	0	1
	PrecStd	76	4	3	6	7	18	33	36	38	19	9	5	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.7	2.4	14.1	24.1	31.5	34.8	36.2	34.2	28.5	19.9	9.7	-3.5
		MCWB	-8.0	-1.6	6.0	11.2	16.0	17.8	21.5	20.9	17.1	10.6	3.9	-5.5
	2%	DB	-10.7	-2.4	10.2	20.2	27.8	31.9	33.1	30.7	25.5	16.2	5.8	-7.7
		MCWB	-11.6	-4.6	4.0	9.3	13.9	17.4	19.4	18.7	15.4	8.3	1.4	-8.9
	5%	DB	-13.5	-5.9	7.1	17.3	25.1	29.7	30.6	28.4	22.8	13.3	2.4	-9.9
		MCWB	-14.3	-7.5	2.0	7.9	13.0	17.1	19.1	17.9	13.9	6.8	-0.8	-10.9
	10%	DB	-16.0	-8.8	4.3	14.5	22.2	27.4	28.5	26.0	20.2	10.5	-0.3	-12.1
		MCWB	-16.6	-9.9	0.2	6.4	11.5	16.4	18.4	17.4	12.7	5.1	-2.5	-12.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.0	-1.5	6.7	12.1	18.0	21.6	25.0	23.1	18.9	11.5	4.1	-5.4
		MCDB	-6.8	2.3	13.2	21.9	27.6	29.9	32.3	30.7	26.0	18.6	9.0	-3.6
	2%	WB	-11.7	-4.6	4.2	9.9	15.4	20.0	22.1	20.6	16.6	8.7	1.5	-8.8
		MCDB	-10.8	-2.4	9.7	19.1	25.4	28.1	29.2	27.4	23.1	15.3	5.4	-7.7
	5%	WB	-14.3	-7.4	2.2	8.2	13.6	18.6	20.6	19.1	14.7	7.0	-0.6	-10.8
		MCDB	-13.6	-5.9	6.6	16.7	23.3	26.2	27.4	25.1	20.9	12.8	2.0	-9.9
	10%	WB	-16.6	-9.9	0.2	6.6	11.9	17.4	19.4	18.0	13.3	5.3	-2.5	-12.8
		MCDB	-16.1	-8.8	4.2	14.0	21.4	24.8	25.8	23.6	19.3	10.2	-0.3	-12.2
Mean Daily Temperature Range	5% DB	MDBR	10.5	13.7	13.7	13.9	14.7	13.5	12.1	12.1	13.6	13.4	11.4	9.9
		MCDBR	12.7	16.4	15.5	18.9	19.4	17.0	16.0	16.1	17.3	18.2	14.1	11.9
		MCWBR	11.8	14.3	10.3	10.8	9.7	6.9	6.3	6.8	8.9	11.1	10.2	10.8
	5% WB	MCDBR	12.6	16.4	15.0	17.5	17.7	14.4	13.0	13.5	15.5	17.4	13.7	11.8
		MCWBR	11.8	14.2	10.1	10.5	9.7	7.1	6.5	6.9	8.4	11.0	10.1	10.7
Clear-Sky Solar Irradiance	taub		0.256	0.280	0.323	0.370	0.405	0.412	0.398	0.376	0.327	0.300	0.276	0.259
	taud		2.268	2.240	2.234	2.206	2.148	2.233	2.327	2.367	2.443	2.447	2.335	2.249
	Ebn at Noon		811	871	885	874	855	850	857	862	877	845	777	748
	Edh at Noon		71	94	114	131	145	134	121	111	92	77	67	64
All-Sky Solar Radiation	RadAvg		1.81	2.97	4.35	5.38	6.23	6.36	5.97	5.18	4.30	2.91	1.82	1.40
	RadStd		0.07	0.12	0.20	0.25	0.43	0.35	0.35	0.36	0.26	0.14	0.09	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	-1.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page